



A comprehensive approach to discover Toxin-Antitoxin systems from human pathogen *Helicobacter pylori*: A poison and its antidote encapsulated in the genome

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ABSTRACT

Aim: An enormous presence and their identified role as stress managers, antibiotic resistance, persistence, and biofilm formation is the reason why the research on Toxin-Antitoxin (TA) loci is getting more and more emphasis. These set of genes consist of poison (Toxin) and its antidote (Antitoxin) expressing in an operon where the toxin inhibits the cellular process and antitoxin which can be a protein or non-coding RNA, rescues the toxin. Most recent progress in genomics and *in silico* studies have revealed new TA families, and types of TA on bacterial chromosome. However, there is almost no or very little is known about the TA in *H. pylori*. Therefore, this study aims to identify the TA genes in human pathogen using a comprehensive *in silico* approach.

Methodology: Here, we have collected the genome-wide data of TA in *H. pylori* 26695 using TASmania, a new TA database. Further, entire TA dataset was validated with several other databases available for TA, operon analysis and experimental data available.

Key findings: The study revealed the presence of 80 putative TA genes in *H. pylori* and highlighted their similarity as well as uniqueness in comparison to other three known TA carrying human pathogens.

Significance: The present study indicates the presence of a large number of TA genes in *H. pylori* which make biofilm and goes into persistence. Hence, our innovative approach unlocks the prospect for characterizing these putative TA genes and their role as stress managers.

1. Introduction

Helicobacter pylori is a gram-negative bacterium which persistently inhabits in the gastric mucosa of human. Its infection is a major problem/risk to the public health. This is a leading cause of mortality because International Agency for Research on Cancer (IARC) has declared it a Class I carcinogen [1,2]. The important concern is the recent rise of multi-drug resistance among the virulent strains that impede eradication of *H. pylori* infection [3]. This bacterium is genetically highly heterogeneous even with the little difference among strains [4]. This can be arisen via variety of mechanisms of genetic material rearrangements either addition or deletion [5] which plays role in pathogenesis and CagPAI pathogenicity island is an example of gene plasticity [6]. A

single reversal mutation in homopolymer sequences for e.g. G to C causes the phase variation at transcriptional and translational level [5]. *H. pylori* has a wide range of virulence gene collection (arsenals) which upsurge the fitness of bacteria to the unfavourable condition and helps in resistance to a broad range of antibiotics, persister cell formation and biofilm formation [7,8]. Persister cells are phenotypically different from normal cells which evade the antibiotic treatment and enter into dormancy [9]. However, there are less identified or more unidentified factors i.e., largely unknown which involved in persistence phenomenon. Several genes/cassettes have been reported that induces this phenomenon such as Toxin-Antitoxin systems (TAs) plays role in the persister cell formation in *Mycobacterium tuberculosis* and *Escherichia coli* [10–12]. Homologs of TAs were discovered onto the bacterial and

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archaeal chromosomes [13–16] after their discovery on plasmids “addiction modules” [17]. Though, their role is a topic of debate however it is reported that they are involved in various physiological processes for instance persistence [18], virulence [19] and biofilm formation [20]. With this, they are now considered to be the stress managers of the cell which helps during stress response and its adaptation in the unfavourable conditions [21–23].

The identification of TAs and their role in persistence of *H. pylori* is not much explored and only four TAs have been characterized so far [24,25]. The majority of TAs remain unidentified and therefore it's important to identify and characterize them for their functional role in the pathogen survival and persistence. These TA cassettes have been divided into quite a few types based upon their product of antitoxin gene and its approach to react, for the reason the product of the toxin gene is a protein in all cases. An antitoxin gene product is small non-coding RNA, in Type I, this stops the translation of toxin [26] and in Type III, it hides away the toxin protein in itself [27]. An antitoxin gene product is exclusively protein in Type II, it inhibits the cognate toxin function by direct binding [28,29] and in Type IV, it inhibits the toxin activity but does not directly binds to the cognate toxin [30].

In this study, we have thoroughly mined and analysed the genome of *H. pylori* 26695 for the identification of TA genes, their families and types. For this purpose, a newly described database for TA identification named TASmania [31] has been explored. Three other known TA carrying human pathogens i.e., *E. coli*, *M. tuberculosis* and *V. cholera* were simultaneously studied to compare and validate all the results identified for *H. pylori*. Several other databases for TA validation and open source operon validation databases were utilized. The study signifies the identification and understanding of the TAs in *H. pylori* which will contribute in designing better strategies to target and develop new therapeutics to treat the infection. Understanding the details of *H. pylori* persistence and its contributing elements will help identify better strategies to treat the infection or may design better drug targets.

2. Methodology

2.1. Identification and data mining of the putative toxin-antitoxin genes of *H. pylori* 26695

The bioinformatic tools, databases, and servers work at more consolidative level to identify, analyse, and categorize the TA genes in different biological networks and pathways which are highly essential to the system biology. In addition to NCBI, here we widely used several other online based platforms such as TADB 2.0 (<https://bioinfo-mml.sjtu.edu.cn/TADB2/index.php>), previously which was a widely used database for TA identification, TASmania (<https://shiny.bioinformatics.unibe.ch/apps/tasmania/>), a new database to identify TA genes, SLING (<https://github.com/ghoresh11/sling/wiki>) a python based Linux programme to identify TA genes, three different operon database i.e. MicrobesOnline Operon Predictions (<http://microbesonline.org/operons/>), Prokaryotic Operon DataBase (PROOpDb) (<http://operons.ibt.unam.mx/OperonPredictor>), and DOOR 2.0 (<http://csbl.bmb.uga.edu/DOOR/>), protein interaction in network models by STRING database (<https://string-db.org/>) and Cytoscape software (<https://cytoscape.org/>) to both collect and validate the data associated with different toxin, antitoxin genes and TA families. We have used these tools to search and identify the putative TA genes in *H. pylori* 26695. Further, we have performed operon-based analysis for the identified putative TA genes and also illustrated TA genes interactomes by using different operon databases and protein-protein interaction analysis.

2.2. Primary data collection

Experimental data associated with TA systems are available for common pathogenic bacteria like *E. coli*, *M. tuberculosis*, *S. aureus* and *V. cholera*. Primarily we have collected the data for *H. pylori* 26695 and

three most common human pathogens (*E. coli* str. K-12 substr. MG1655, *M. tuberculosis* H37Rv, *V. cholerae* O1 biovar El Tor str. N16961) as reference to represent the proof for the entire dataset and TA validation. The reference whole genome as well as proteome sequence and putative toxin and antitoxin nucleotide and protein sequences and annotations of *H. pylori* 26695 chromosome were obtained from NCBI with accession number of NC_000915.1.

Here, we have used toxin-antitoxin database 2.0 (TADB 2.0) (<https://bioinfo-mml.sjtu.edu.cn/TADB2/index.php>) [32] which is a resource of type-II TA loci of over 900 prokaryotic genomes based on PostgreSQL system. TADB 2.0 is available since 2010 and lastly updated in 2017 and an additional software is also available called TAFinder. TAFinder is a web-based tool to identify Type II toxin-antitoxin loci in bacterial genome which is free and open to all academic users. But the data recovered for *H. pylori* in both TADB 2.0 and TAFinder is comparatively very less than the expected presence of TA loci in this persister former pathogen. Therefore, for our studies we found a new open-source database TASmania available at <https://shiny.bioinformatics.unibe.ch/apps/tasmania/> which is the latest TA database for *in silico* identification of toxin-antitoxin in bacterial genomes. It is based on Ensembl bacteria database and helps in toxin-antitoxin and TA combination analysis by clustering the toxin-antitoxin Hidden Markov Models (HMMs) profiles. By using TASmania database we had collected data (Ensembl gene id, Start and End point of genes, on which strand they locate, Protein Length, Gene Description, Operon Id, hmm_E value, TAS Info, Operon Structure, Tasmania hmm cluster Id and pfam desc) for the bacterial genomes at different *E*-values from 0.02 to 1e-100. It includes a webtool named TASer (<https://shiny.bioinformatics.unibe.ch/apps/taser/>) which utilizes the HMM profiles from TASmania database and a High-Performance Computing Cluster to locate the putative toxins and antitoxins [31].

2.3. Secondary data collection

Though putative toxin-antitoxin genes were collected from TASmania and TASer, very few genes are having experimental evidence in case of *H. pylori* 26695 and these evidences suggested that a pair of TA genes involved in a biological activity are encoded in a single operon and commonly transcribed from same promoter [24,25]. So, we have used an open-source command line tool, SLING (<https://github.com/ghoresh11/sling/wiki>) which is implemented in Python (version 2.7). SLING uses the assembled contigs or complete genomes of bacteria as input source and performs search for the closest TA genes with no limitations [33]. For this purpose, we used the FASTA format of complete genome of *H. pylori* 26695 to find the longest coding sequence (CDS) and GFF format used for the annotation files of the genome to increase the search sensitivity. Furthermore, we know that many important gene systems are encoded in an operons and TA system is one of them [39]. We selected *H. pylori* 26695 genome as an input in three different widely used open source operon database MicrobesOnline Operon Predictions (<http://microbesonline.org/operons/>) [34,40], Prokaryotic Operon DataBase (PROOpDb) (<http://operons.ibt.unam.mx/OperonPredictor>) [35] and DOOR 2.0 (<http://csbl.bmb.uga.edu/DOOR/>) [36] to study the TA operons in *H. pylori* 26695. We collected the data available for each putative TA gene and analysed their position in their respective operon models.

To get an overall knowledge about sequence alignment analysis for the putative TA genes identified by TASmania database for *H. pylori* 26695, we performed NCBI protein-protein BLAST (Blast-p) (<https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins>) in two different conditions. Initially, we took all putative TA genes of *H. pylori* 26695, identified by TASmania without using any *E*-value filter and performed Blast-p against the whole proteome of three common human pathogens and secondly, we performed Blast-p against only the experimentally reported toxin-antitoxin sequences in three other known TA carrying pathogens. All the Blast-p results were tabulated, and different chart

models prepared for quick and easy analysis.

2.4. Network based analysis of putative toxin-antitoxin genes

Interaction study of putative TA genes of *H. pylori* 26695 can be used to find the associated toxins to their antitoxins and vice versa and their associated other genes which are playing important role in cell signaling pathways, pathogenesis, and persistence. We used an online based classification system i.e. STRING (<https://string-db.org/>) which implements KEGG and Gene Ontology classification system [37]. We performed individual input for each putative TA genes in STRING to find out the network-based protein-protein interaction and predict the closely relative proteins and TA genes. We used an open-source software platform Cytoscape (<https://cytoscape.org/>) [38] to prepare and visualize the interactome for all putative TA genes of *H. pylori* 26695. A flowchart is representing the overview of methodology used for the identification and comparative analysis of putative TA genes in *H. pylori* 26695 (Fig. 1).

3. Results

3.1. *H. pylori* 26695 may have a high number of putative TA systems

Manual data mining of *H. pylori* 26695 chromosome accession no. NC_000915 (1.68 Mb genome) revealed only four TA gene sets which has been experimentally proved. In addition to our genome of interest i.e. *H. pylori*, we have taken three other most common human pathogens carrying enormous number of TA systems as our search inputs in Toxin-Antitoxin database 2.0 (TADB 2.0) [32], which is considered as an updated type-II TA database published in 2018. This database is widely used to find TA genes, as it gives information for both *in silico* predicted and experimentally proved TA genes with their TA locus and genetic features. As an exceptional and useful TA database, it has been cited by 118 research groups. When we used *E. coli* str. K-12 substr. MG1655 (Tax_id: 511145) as search input, we found total 48 numbers of TA genes

out of which 39 are experimentally proved and 28 are type-II TA genes. While, in case of *V. cholerae* O1 biovar El Tor str. N16961 (Tax_id: 243277) we found 32 type-II TA genes where all of them were experimentally proved. In the field of TA system, the most explored pathogen is *M. tuberculosis* [10]. When we used *M. tuberculosis* H37Rv (Tax_id: 83332) as search input, we found a total no. of 152 type-II TA genes out of them 88 are experimentally proved. But when we searched for TA genes in *H. pylori* 26695 (Tax_id: 85962), we found only 4 type-II TA genes where HP0892 – HP0893 and HP0894 – HP0895 are 2 TA pairs, which is the lowest among these four common human pathogens. As described by Xie, Wei et al. 2018 we also used the TA finder which is associated with TADB 2.0. We had used *H. pylori* genome, accession Number NC_000915 as an input to find TA genes present in *H. pylori*, a gastric human pathogen and here we found the same results as revealed in TADB 2.0. Therefore, instead of following a single database to find TA genes, we explored for other databases/tools in literature and found TASmania database (<https://shiny.bioinformatics.unibe.ch/apps/tasmania/>) [41] which has already been used to show the TA genes in *M. tuberculosis*.

3.2. Comparative analysis of putative TA genes identified in *H. pylori* 26695 and other known TA carrying organisms by TASmania database

TASmania is the latest open-source database freely available for *in silico* identification of TA genes in bacterial genomes. It offers a vast repertoire of annotated assemblies of TA loci in different bacterial genomes in a very presided manner which can be searched by the organism's name and can be filtered by different *E*-values. Here, in this study, total 80 TA genes (without using filter) were identified using TASmania., and in case of *E. coli* str. K-12 substr. MG1655, 135, in *V. cholerae* O1 biovar El Tor str. N16961, 111 and in *M. tuberculosis* H37Rv the number obtained were 269 which is the largest among the four most common pathogens as shown in Fig. 2A. When the data resulted was compared from TASmania and TADB 2.0, it was found that TASmania showing comparatively high number of putative TA genes. Importantly, 28% in

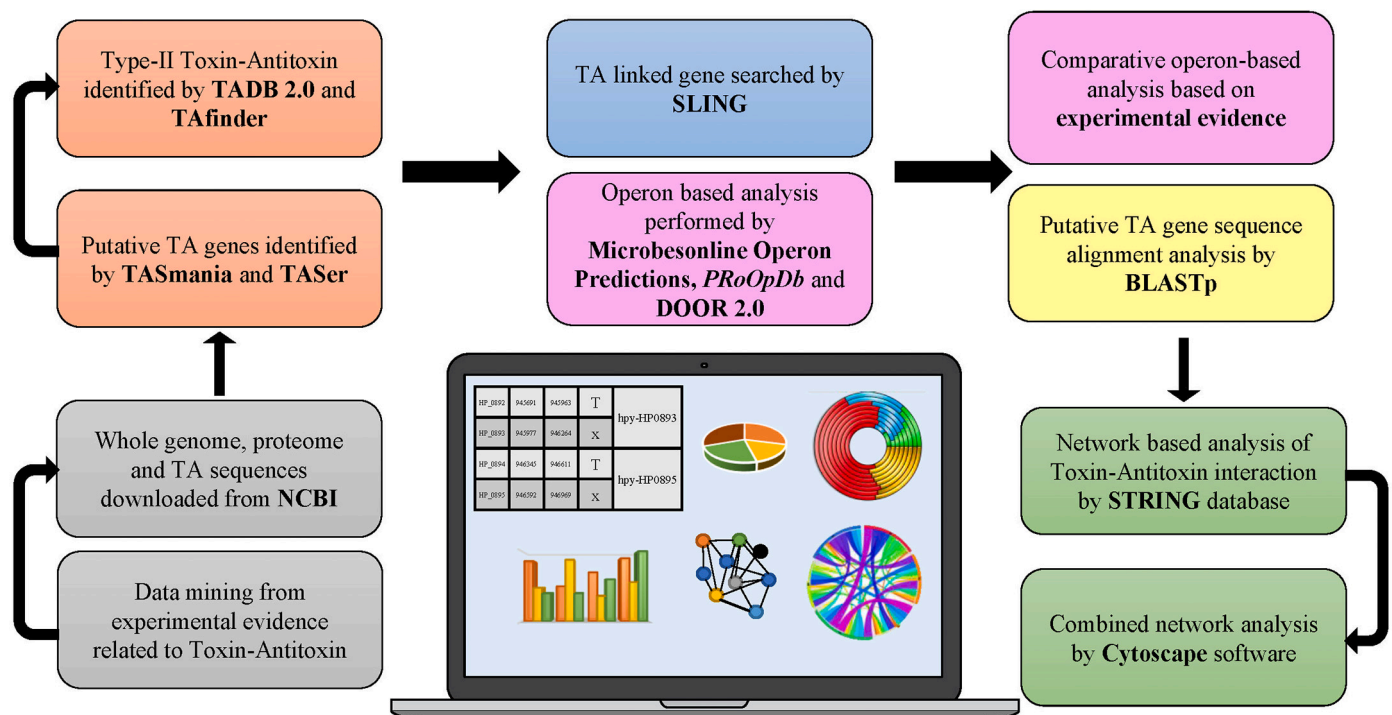
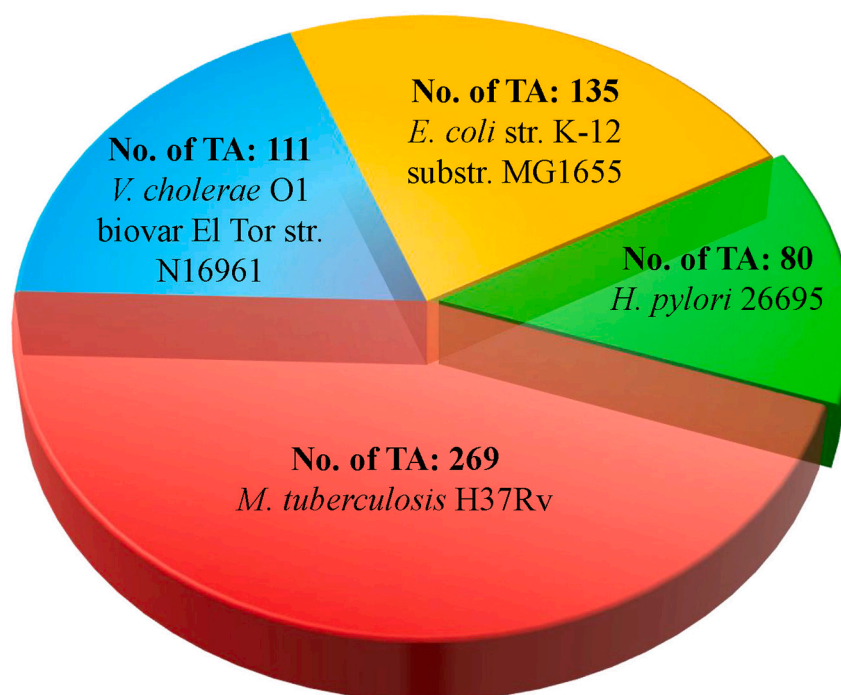
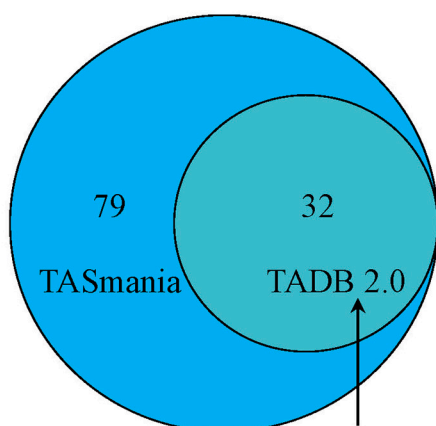


Fig. 1. Overview of identification and comparative analysis of putative TA genes in *H. pylori* 26695. Different steps which involves in data mining from different databases and servers, identification of toxin-antitoxin genes present in *H. pylori* by TADB 2.0, TAFinder, TASmania and TASer, Operon models analysis by Microbesonline Operon Predictions, PROpDb and DOOR 2.0, sequence-based analysis by BLASTp and interaction analysis by STRING and Cytoscape.

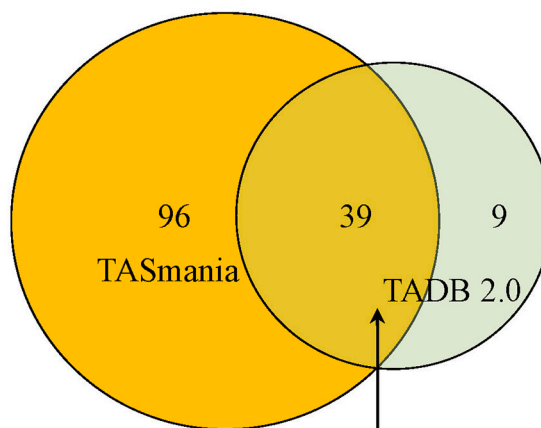
(A)



(B)

(i). *V. cholerae*

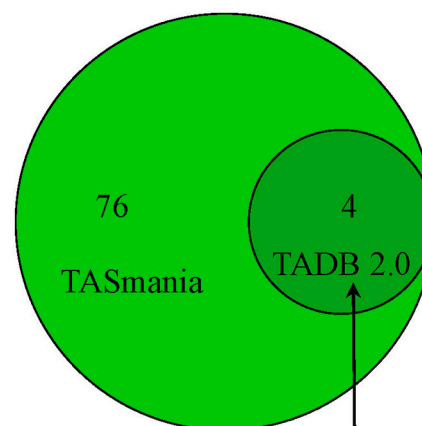
Experimentally proved Type-II TA (32)

(ii). *E. coli*

Experimentally proved Type-II TA (39)

(iii). *M. tuberculosis*

Experimentally proved Type-II TA (88)

(iv). *H. pylori* 26695

Experimentally proved Type-II TA (4)

Fig. 2. (A) Comparative chart of TA identified for *H. pylori* and other known TA carrying organisms by TASmania without filter (B) Comparison of TASmania and TADB 2.0 hits with the experimentally validated Type-II TA (i) *V. cholerae* O1 biovar El Tor str. N16961, (ii) *E. coli* str. K-12 substr. MG1655, (iii) *M. tuberculosis* H37Rv and (iv) *H. pylori* 26695.

E. coli, 28% in *V. cholerae* and 32% in *M. tuberculosis* experimentally identified and validated TA genes are also listed in TASmania (Fig. 2B) which provides valuable clues in identification and understanding the TA system and their families in the human pathogen *H. pylori*.

Another important aspect which TASmania provided here i.e., enormous information related to the putatively mined TA which portrays a resource of key importance. This includes Ensembl gene id which represents the ORFs, Start and Endpoint of genes i.e. location of the gene on the chromosome which has been used to calculate the length of the toxin/antitoxin gene and find out whether there is overlap present between the corresponding gene or presence of the intergenic gaps, presence of TA genes on the coding or non-coding strand, length of the protein, gene description represents the associated molecular functions, Operon Id represents pseudo-operons, hmm_E_value shows the HMM profiles into larger HMM clusters by similarity, TAS info shows the type of TA, operon structure shows the position of toxin and antitoxin in a single operon, Tasmania hmm cluster Id helps to keep a link between HMM profiles and their respective HMM clusters and Pfam desc shows the nearest Pfam families for putative TA genes. All this data was collected for the query genome i.e., *H. pylori* 26695 and other three known TA carrying pathogens at different E-value filters such as 0.02, 0.001, 1e-04, 1e-05, 1e-10, 1e-20, 1e-30, 1e-50, 1e-100 and no E value filter was also applied. Data obtained from the no E-value filter which was representing the total number of putative TA genes identified has been represented in Fig. 3A. By applying the E-value filter from 0.02 to 1e-100, number of putatively identified TA genes were obtained in a highest to lowest range i.e., highest at 0.02 and lowest at 1e-100 in case of all four organisms as shown in Fig. 3A and list of name of genes at different filters is provided in Supplementary Table S1. All the 80 TA genes of *H. pylori*, identified at no E-filter have been represented with their gene name on the circular genome map (Fig. 3B).

As of now, only four TA genes i.e., two pairs of TA have been verified and characterized experimentally in case of *H. pylori* 26695 (HP0892 – HP0893 and HP0894 – HP0895) out of 80, identified by TASmania database (shown in orange colour in Fig. 3B). In HP0892 – HP0893 TA pair, HP0892 is a YafQ-homologous toxin and HP0893 is an antitoxin where the toxin inhibits RNase activity and cell toxicity of the toxin HP0892 when binds to the antitoxin HP0893 [25]. In HP0894 – HP0895, HP0894 acts as a toxin with intrinsic RNase activity, and it has a high structural resemblance with RelE, and YoeB toxins. HP0895 functions as an antitoxin by binding to both DNA and toxin HP0894 and helps in neutralizing the toxicity of HP0894 [24].

3.3. Pfam analysis of putatively identified TA genes of *H. pylori* 26695

The Pfam database is one of the most important protein data libraries for classifying the proteins and their families, which is based upon multiple sequence alignments and HMM profiles. TASmania database provides the Pfam description for putative TA genes if available. In *H. pylori* 26695, total identified putative TA genes were covering the 17 Pfam families namely AbiEii, CptA_toxin, DUF1015, Fic_N, HicA_toxin, HicB_lik_antitox, HipA_C, HOK_GEF, MraZ, NikR_C, ParBc, PhdYeF-M_antitox, PIN_5, SpoIIISA_toxin, SymE_toxin, YafQ_toxin and Zeta_toxin. Further, to confirm the presence of the obtained Pfam families for *H. pylori* with other known TA families in three TA carrying pathogens were studied. Interestingly, 12 families similar to *H. pylori* were present in *E. coli*, 5 in *M. tuberculosis* and 3 in *V. cholera* while 5 out of 17 were found unique to *H. pylori* 26695 as shown in the Fig. 4.

3.3.1. AbiEii

AbiEii family was identified in *E. coli* and *M. tuberculosis* which

belongs to the nucleotidyltransferase superfamily (Table 1). This protein family induces bacteriostasis and helps in triggering the TA system upon infection with phage to terminate the cellular activities. These phage resistance systems are working as novel type-IV TAs in bacteria [42,43].

3.3.2. CptA_toxin

CptA toxin family consist of bacterial proteins including inner membrane protein YgfX from *E. coli* which is previously known as CptA (Table 1). Primarily, this family was considered as a toxin family but experimentally found that it does not affect cell growth. It showed interaction with MreB and FtsZ, two essential cytoskeletal proteins in *E. coli* and found to be responsible for the YgfX toxicity to block the polymerization of FtsZ and MreB, it inhibits FtsZ which is GTP-dependent and MreB is ATP-dependent polymerization [44].

3.3.3. HicA_toxin

HicA_toxin family was identified in *E. coli* and *V. cholerae* which characterises the HicA mRNA interferase family (Table 1). This toxin family forms the toxic component of a TA module. The proteins belong to this family contain a double-stranded RNA-binding domain which acts as a translation-independent mRNA interferases [45–47].

3.3.4. HicB_lik_antitox

HicB-like antitoxins family was reported in *E. coli* TA systems and this protein family is associated with a novel TA system that contains a partial RNaseH fold or ribbon-helix domain (Table 1). The proteins belong to this family forms a stable DNA binding module and interacts in an established way with HicA and plays an important role in RNA-binding and may also in cleavage [45,48,49].

3.3.5. HipA_C

The protein members of HipA_C family is equal to a region adjacent to the C-terminus of the HipA proteins and found to be expressed in all the three pathogens i.e. *E. coli*, *M. tuberculosis* and *V. cholerae* (Table 1). It is reported that the proteins are identified as they have involved in high-frequency persistence to the toxic effects of inhibition of either DNA or peptidoglycan synthesis. HipA_C proteins are strongly connected with HipB to form the HipA-HipB complex which may support the autoregulation of hip operon but when expressed alone, it is lethal to bacterial cells. Experimental evidence suggests that these hip proteins possess control on cell division by interacting with cell division genes or their products. This protein family is also found in the serine/threonine-protein kinase CtkA of *H. pylori* which plays an important role in the induction of host inflammation by inducing the release of cytokines [50–54].

3.3.6. HOK_GEF

The proteins belong to the hok/gef family is found to be toxic to the cells when overexpressed and this is a common mechanism seen in gram negative bacteria. Some of these proteins play a crucial role in RNA stability, some induces the degradation of stable RNA at high temperature and other proteins of the family disrupt the synthesis of cellular magnesium. In *E. coli*, these proteins are reported to play roles associated with intracellular toxicity (Table 1) [55,56].

3.3.7. MraZ

This family is highly conserved in bacteria and the proteins belong to it are essential for bacterial cell growth and helps in survival of the organism at extreme environment by playing a crucial role in cell division and cell-wall biosynthesis. Further, the proteins of MraZ antitoxin family, shows structural similarities to AbrB transition state regulator

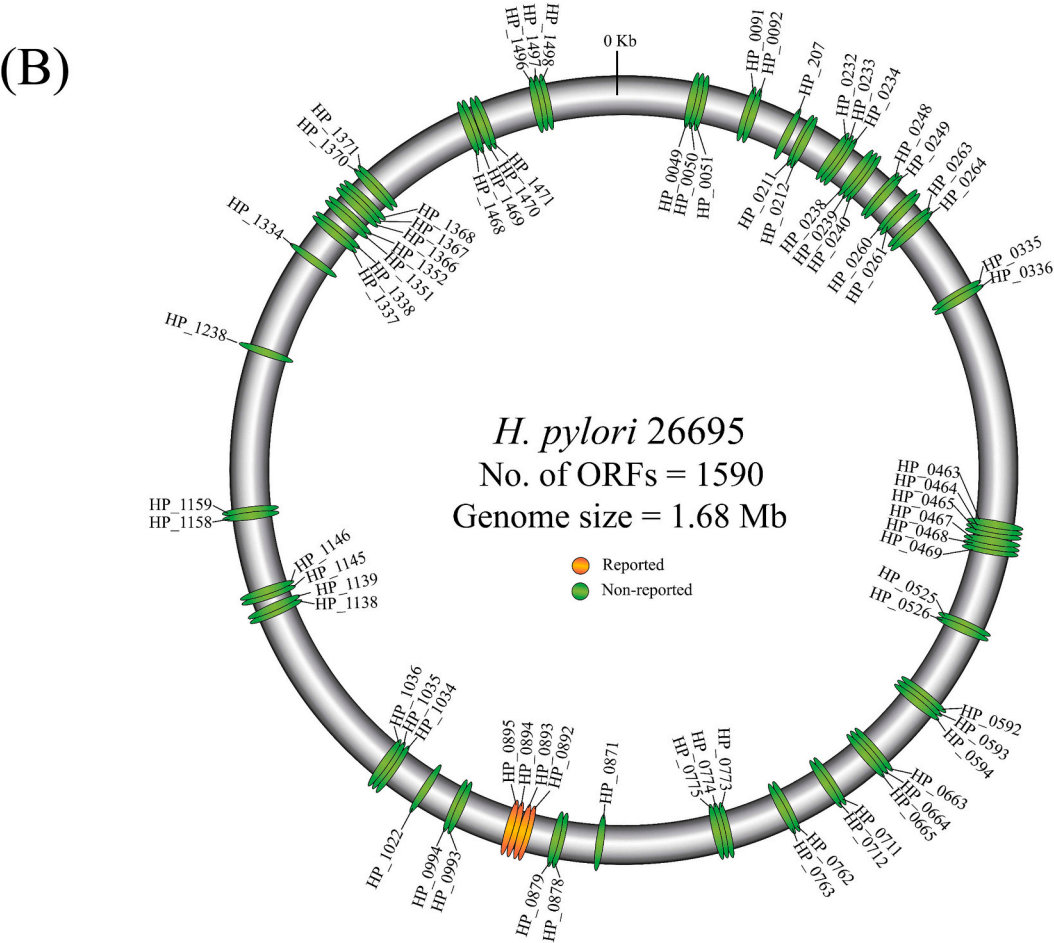
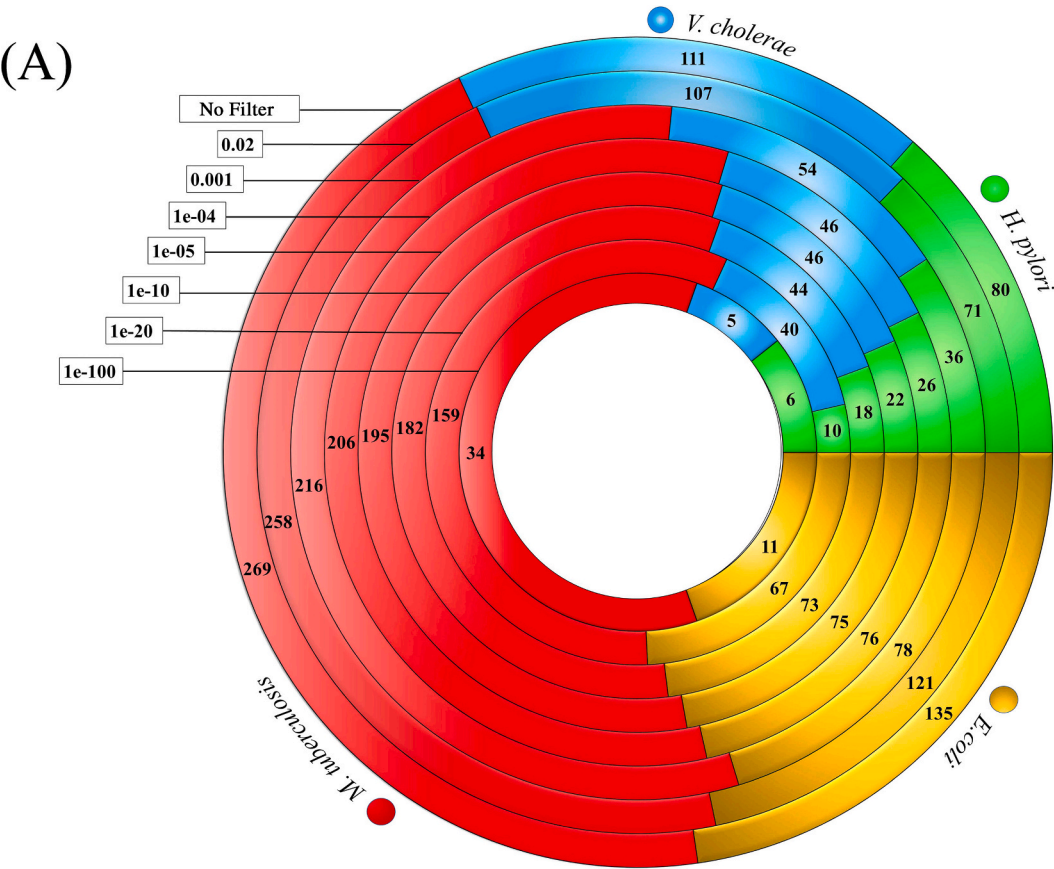


Fig. 3. (A) Comparative chart of TA identified by TASmania for *H. pylori* and other known TA carrying organism at different E-value (No filter, 0.02, 0.001, 1e-04, 1e-05, 1e-10, 1e-20, 1e-30, 1e-50, and 1e-100) (B) Circular genome representing all 80 putative TA genes in *H. pylori* 26695 (orange colour representing experimentally validated/reported and green colour representing non-reported TA genes). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

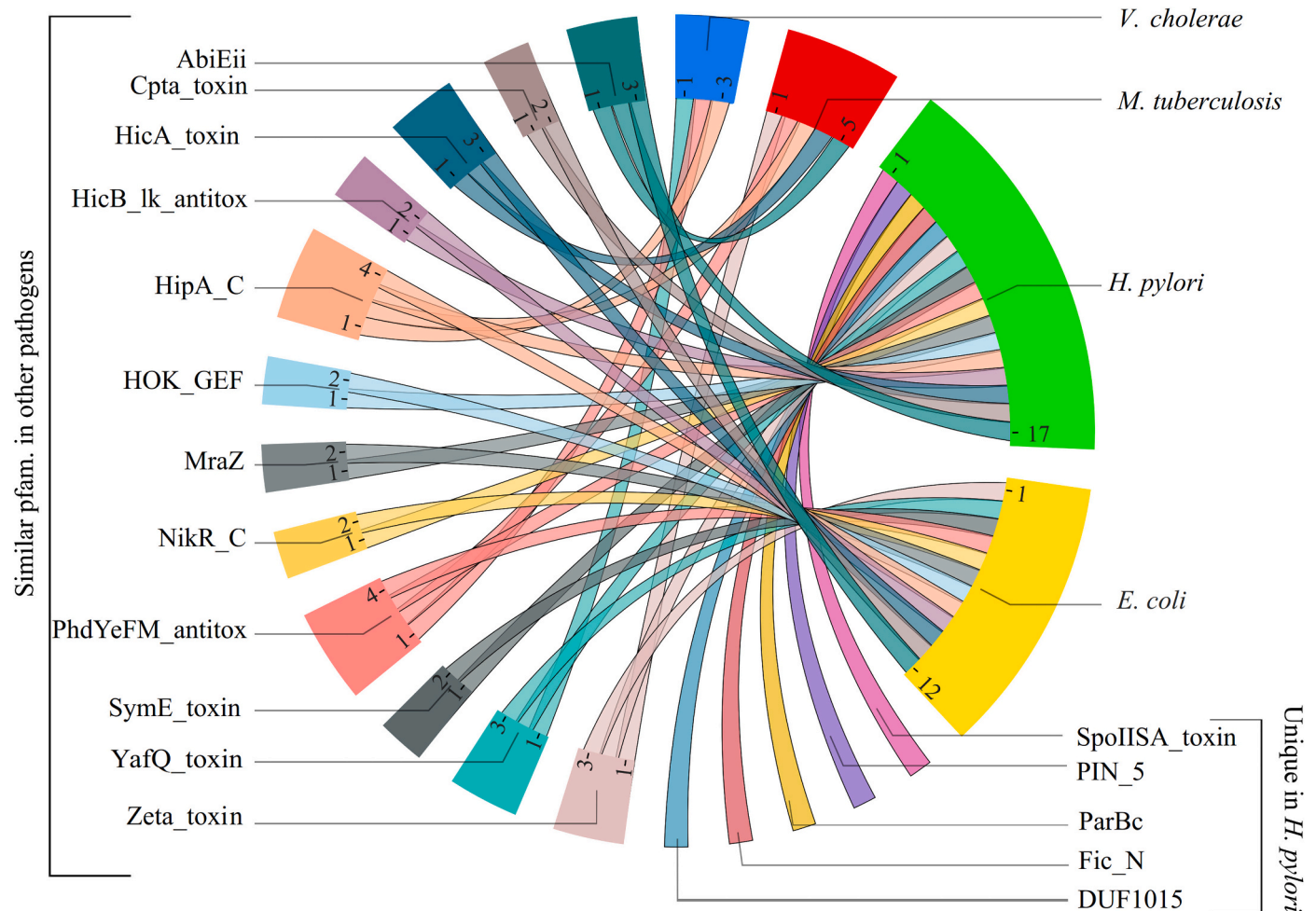


Fig. 4. Schematic representation of predicted *H. pylori* Pfam Toxin-Antitoxin families with three known human pathogens.

and functional similarities with other different antitoxins and act as DNA binding transcription factors. As per the literature, in *E. coli* MraZ family proteins acts as a transcriptional regulator (Table 1) [57,58].

3.3.8. NikR_C

This family is the only known metal responsive transcription factor in pathogenic bacteria and belongs to ribbon helix-helix superfamily. These proteins could not show any kind of detectable DNA binding mechanism in the absence of nickel. Regulation of nickel uptake is important because the excess nickel is toxic for cells. It is reported that in *E. coli* the concentration of nickel metal is controlled by the transcriptional repression of nickel specific ATP-binding cassette (ABC transporter complex) which is encoded by the *nikABCDE* operon (Table 1) [59,60].

3.3.9. PhdYeFM_antitoxin

The proteins of this family plays role as antitoxins in type II toxin-antitoxin systems which is found in *E. coli*, *M. tuberculosis* and *V. cholera* (Table 1). These proteins can bind to their cognate toxins and suppress the TA expression by binding to the genomic DNA via N-terminus. This antitoxin protein and the TA complex formed can undergo unfolding as the temperature rises [61–63].

3.3.10. SymE_toxin

Both type-I and type-II toxins are found in this family. SymE (SOS-induced *yjiW* gene similar to MazE) is a SOS-induced toxin which inhibits the cell growth by enhancing the RNA degradation. The SymE_toxin found in *E. coli* has structural homology with MazE family, which is a transcriptional factor and serve as antitoxin in various type-II TA modules by cleaving mRNA (Table 1). Though, SymE toxin also belong to type-I TA system, it does not show any functional homology to other type-I TA [64].

3.3.11. YafQ_toxin

YafQ toxin family serves as toxin-antitoxin system in *E. coli*, *V. cholerae* and also reported in *H. pylori* which is a ribosome-associated endoribonuclease (Table 1). It associates with the ribosome and blocks translation elongation through sequence-specific and frame-dependent mRNA cleavage [46,65,66]. It is reported that both experimentally identified toxins i.e., HP0892 and HP0894 present in *H. pylori* 26695 belongs to this toxin family [24,25].

3.3.12. Zeta_toxin

The toxin proteins belong to Zeta toxin are found in pathogens and play a pivotal role in the inhibition of intracellular metabolism. These

Table 1

Similarity of predicted *Hp* Toxin-Antitoxin families with their experimental evidence in three known pathogens.

TA family of <i>H. pylori</i> 26695	TA type	<i>E. coli</i> str. K-12 substr. MG1655	<i>V. cholerae</i>	<i>M. tuberculosis</i> H37Rv
AbiEii	IV	Dy, Przybilski et al. 2014	×	Beck, Usher et al. 2020
Cpta_toxin	IV	Masuda, Tan et al. 2012	×	×
DUF1015		×	×	×
Fic_N		×	×	×
HicA_toxin	II	Jørgensen, Pandey et al. 2009, Turnbull and Gerdes 2017	×	Bordes, Sala et al. 2016
HicB_1k_antitox	II	Jørgensen, Pandey et al. 2009, Turnbull and Gerdes 2017, Manav, Turnbull et al. 2019	×	×
HipA_C	II	Moyed and Broderick 1986, Black, Kelly et al. 1991, Black, Irwin et al. 1994, Golub, Panzer et al. 1988, Poulsen, Refn et al. 1991, Adams, Udell et al. 2005, Eraso, Markillie et al. 2014, Schreiter, Sintchak et al. 2003, Wang, Dias et al. 2004	Paranjape and Shashidhar 2019	Singh, Barry et al. 2010
HOK_GEF	I	×	×	×
MraZ		×	×	×
NikR_C		×	×	×
ParBc		×	×	×
PhdYeFM_antitox	II	Nieto, Cherny et al. 2007	Iqbal, Guérout et al. 2015	Roy, Kundu et al. 2019
PIN_5		×	×	×
SpoIIISA_toxin		×	×	×
SymE_toxin	I	Kawano 2012	×	×
YafQ_toxin	II	Motiejūnaitė, Armalytė et al. 2007, Kolodkin-Gal, Verdiger et al. 2009	Hadži, Garcia-Pino et al. 2017	×
Zeta_toxin	II	Mutschler, Gebhardt et al. 2011, Rocker and Meinhardt 2015	×	Jaén-Luchoro, Aliaga-Lozano et al. 2017

toxin proteins are thought to be a part of postsegregational killing (PSK) due to their toxic nature. Several proteins namely polynucleotide kinase and 2',3'-cyclic-nucleotide 3'-phosphodiesterase also belongs to this family. In *E. coli* and *V. cholerae*, the zeta toxin operates as a kinase domain in respective TA modules (Table 1) [67–69].

MraZ and NikR_C, are two Pfam families, found in *E. coli* [57,58,59,60] but there is no experimental evidence which can define the type of TA genes associated with these families. The other five Pfam families which includes DUF1015, Fic_N, ParBc, PIN_5 and SpoIIISA_toxin are found to be unique in *H. pylori* 26695 (Fig. 4) as there are no experimental evidence related to these families and their function as TA families in other three pathogens. However, after searching the literature related to the unique *H. pylori* 26695 Pfam families it was found that SpoIIISA_toxin is a TA family belongs to Type II in *B. subtilis* [70] and Fic_N contains the sequence similarity with the RelE family of TA which belongs to Type II [71–73].

3.4. Validation of TASmania identified TA genes using SLING

To validate the putatively identified *H. pylori* 26695 TA, we have used a tool named SLING (<https://github.com/ghoresh11/sling/wiki>), implemented in python 2.7 which is used for the Search of Linked Genes. This is known for identification of two types of linked genes but not limited to i.e., Toxin-Antitoxin and RND efflux pumps which presents as an operon. This tool predicted the toxin gene based upon Hidden Markov Model algorithm followed by recognition of its corresponding linked gene which is antitoxin gene. At next step, SLING identified the clusters of TA system homologs using BLAST clustering. SLING requires the complete genome or a collection of assembled genomes and contigs, HMM profiles of the multiple sequence alignments of proteins and GFF format annotation files of a genome to search the homologs proteins [33]. This tool used the default structural parameters for TA filtering, where the minimum and maximum toxin length fixed between 30 aa to 200 aa, the minimum and maximum antitoxin length fixed between 50 aa to 150 aa and the position of antitoxin either upstream or downstream to the toxin which can be changed manually. Using SLING, we were able to identify the TA genes and the neighbouring TA pairs and alongside this tool represents the number of existences of each hit, partner genes, complete array groups and discarded hits found in *H. pylori* 26695 genome. The output files (Supplementary Table S2) represent the domain of TA genes, HMMER score, hit length followed by hit start and hit stop regions and the discarded hits stated as excluded ORFs. Out of 80 putative TA of *H. pylori*, only 22 are found in the SLING database which includes all the 4 previously reported TA genes.

3.5. Operon based analysis of TAs in different operon databases

The command-based tool SLING can give the putative TA genes with their neighbouring pairs but not the TA genes position in an operon. So, an exhaustive operon-based analysis was needed for the putatively identified TA genes of *H. pylori* 26695. We have used three different operon database i.e. MicrobesOnline Operon Predictions (<http://microbesonline.org/operons/>) [34,40], Prokaryotic Operon DataBase (ProOpDB) (<http://operons.ibt.unam.mx/OperonPredictor/>) [35] and DOOR 2.0 (<http://csbl.bmb.uga.edu/DOOR/>) [36] to do the comparative analysis of all 80 putative TA genes and their position in their respective operon models.

3.5.1. Operon based analysis by MicrobesOnline Operon predictions database

In 2005, Price, Huang et al. 2005 reported that in genome of human pathogen *H. pylori* very less operons are known however it is expected that there may be some large number of operons present but not observed yet. Further, the same group designed a database MicrobesOnline Operon Predictions where they surveyed the operon structures in 124 genomes. The updated database is having the operon models for 1335 genomes. This database was considered as best for the distance between predictions and genome-specific microarrays. In our studies, we simply selected the *H. pylori* 26695 from the 1335 genome list and this database gave the operon models with the consisting gene description, systematic name for individual genes present in a single pair, bOp (whether predicted pair lie in the same or different operons by mentioning as TRUE or FALSE), pOp (the estimated probability if the gene pairs are in the same operon when the value is close to 0 or 1 the prediction stated as confident and near to 0.5 stated as low confidence predictions), Sep (shows the distance between the predicted two genes in base pairs), two different scores MOGScores (the conservation range from 0 to 1, where 0 stands for not conserved and 1 stands for 100% conserved), and GOScore (showed the fraction of genome if one of the genes in operon is missing or not characterized), COGSim (the predicted genes in the operon are belonging to same COG category or not) and ExpSim (the score based on the correlation of expression patterns for the predicted operons). We found total 1479 genes of *H. pylori* 26695 in the

same or different operon models by Microbesonline Operon Predictions database. Out of 80 putative TA genes identified by TASmania, 78 genes are predicted to be in the same or different operon models two genes i.e. HP0207 and HP1370 are not present in any operon model. In case of *H. pylori* genome, the scores recorded for distance was 0.275 while in *E. coli* it was 0.406. Total 87 operon models are predicted for the 78 putative TA genes and out of 78 genes, 53 genes are involved in multiple operons (Supplementary Table S3). However, the limitation of this database is not to deduce the exact position of the putative TA genes in an operon model.

3.5.2. Operon based analysis by Prokaryotic Operon DataBase

Here, we used another repository of operon predictions i.e. The Prokaryotic Operon DataBase (ProOpDB). This database works on highly accurate operon identification algorithms and a precompiled phylogenetic distance matrix, it includes 25,49,412 predicted operons of more than 1200 prokaryotes with an extra feature by including the RNA genes [35]. This database offers various alternatives by which the total sets of predicted operon models can be retrieved including the name of the prokaryotes, known metabolic pathways, gene ontology (defined by COG database), conserved protein domains, and reference genes with their respective operons. ProOpDB is working on five dependencies like KEGG flat-file database, COG database, the *hmmsearch* program HMMER package, BLAST comparisons in bi-directional best hits (BDBH) and lastly on STRING database. By selecting the genome *H. pylori* 26695 from the selection list as an input we got our results for operon models for complete genome with individual operon names (hpy used as initials for predicted operons sheet generated by ProOpDB), the genes present in the operon with their positions in the operon model and genome and the description of the genes. For *H. pylori* 26695 this database shown total 1520 genes present in 721 operons, out of 80 putative TA genes identified by TASmania, 76 genes present in 47 predicted operon models and four genes i.e., HP0335, HP0712, HP1146, and HP_1370 are excluded by ProOpDB (Supplementary Table S4).

3.5.3. Operon based analysis by DOOR 2.0 database

The above mentioned two databases differ in their coverage of the predicted operon information but they are lacking experimental validation and single or multi-gene operon, models. This DOOR 2.0 database, which consists of 2,45,685 operons for 2072 prokaryotic organisms and 1645 plasmids out of which 1385 are experimentally verified, used for our studies. 13,23,902 multigene operons, 25,78,949 single gene operons, 69,75,454 conserve operons with 34,56,718 *Rho-independent* terminators for 2072 prokaryotes. In addition to this, DOOR 2.0 is based on NCBI genome FTP server which consists of 133 archaea and 1939 bacteria. In this database, the operon-based data can be predicted by uploading three types of data including chromosomal DNA sequence, protein sequence and gene location. we can select the specific organism from the list to find the operons in the genome by using the web-based tool. We used *H. pylori* 26695 as an input to get the predicted operon models for the entire genome and the output results which includes Operon id (which is same for the individual genes present in the same operon), start and end position in the genome with their amino acid length, on which strand they locate, respective COG number if available, their description and displayed both the single gene and multi-gene operons. In our studies, we found total 1573 genes for *H. pylori* 26695 belongs to 703 operons. Out of 80 putative TA genes, 70 genes present in 32 operons where HP0212, HP0264, HP0594, and HP1371 are present alone in single gene operons and 28 operons are multi-gene operon models. The putative TA genes namely HP0207, HP0211, HP0263, HP0871, HP1022, HP1238, HP1334, HP1370, HP1468, and HP1469 are excluded by DOOR 2.0 as they are not present in single or multi-gene operons (Supplementary Table S5).

3.5.4. Comparative operon-based analysis based on experimental evidence

To verify our operon-based analysis we did a comparative study with

the operon-based data available by Sharma, Hoffmann et al. 2010 as they experimentally verified it. They had performed mapping and annotation of the primary transcriptomes for *H. pylori* and presented the operons and transcriptional start sites (TSS) by performing a novel approach i.e. differential RNA sequencing (dRNA-seq) [74]. The transcriptome of *H. pylori* is highly complex, and they have found that 27% of the primary TSS are antisense TSS. So, it is highly necessary to know whether the putative TA genes are positioned on the sense strand or on the non-sense strand in a *Hp* bacterial genome. We plotted all the 80 putative TA genes with their protein length and found a 1:1 ratio for upstream and downstream (Fig. 5A). Out of 80 TASmania identified TA genes only five genes HP0207, HP0871, HP1022, HP1334 and HP1370 were not present in Sharma et al., datasets. The remaining 75 putative TA genes are present in 36 operon models which are in matching with the data presented in our study by DOOR 2.0 operon models. They have also provided the list of identified genes with their position in an operon or in multiple operons, so we have matched the similarity of the classification of operon of our putative genes with their operon classifications. These operons are grouped in seven categories as per Sharma et al., 2010, where we found operons from our study classifies in the following categories i.e., 11 in Alternative, 23 in confirmed, 8 in extended, 4 in single, 2 newly identified, 1 in confirmed/alternative, 1 in ambiguous and 0 in dismissed. Further, we have matched the categories of the 75 putative TA genes according to their operon classification and found 17 alternative, 50 confirmed, 14 extended, 4 single, 3 newly identified, 2 confirmed/alternative, 2 ambiguous and 0 dismissed. According to the comparative operon-based analysis, we got the similar results which strengthened our collected data (Supplementary Table S6). Based on our operon analyses for the putative TA genes there are 42 T-A and A-T positioned TA pairs and 18 TA pairs are found to be upstream regulated and the remaining are downstream-regulated. In downstream-regulated TA pairs, T-A positioned gene pairs are high in number, but A-T positioned TA pairs are less in number than the upstream regulated TA pairs (Fig. 5B).

3.6. Sequence alignment analysis of putatively identified *H. pylori* 26695 TA genes against other TA carrying pathogens

Here, we exhaustively analysed the sequence alignment using BLASTP for all the putatively identified TA genes by TASmania database for *H. pylori* 26695 in two different conditions. Initially, we took all 80 putative TA protein sequences of *H. pylori* as 'Query Sequence' against the 10371 microbial proteins of *E. coli* str. K-12 substr. MG1655 (taxid:511145), 10824 microbial proteins of *M. tuberculosis* H37Rv (taxid:83332) and 6510 microbial proteins of *V. cholerae* O1 biovar El Tor str. N16961 (taxid:243277) one by one and default algorithm parameters at 0.05 expected threshold was selected. All the BLASTP results were analysed based on the output of gene description, max score, query cover, *E*-value, and percentage identity. Then, we prepared our data-sheet by looking into the four major characteristics i.e., query cover, *E*-value, and percentage identity where we filtered out the BLASTP results by selecting the query cover more than 50%. Here we found, HP0892 and HP0894 the reported toxin genes of *H. pylori* showing 96% and 56% query cover respectively with the ribosomal dependent mRNA interference toxin YafQ of *E. coli* str. K-12 substr. MG1655 while in *V. cholerae* O1 biovar El Tor str. N16961 the query cover found for both HP0892 and HP0894 were 100% but no matching was found with *M. tuberculosis* H37Rv (Supplementary Table S7). In Fig. 6 (A, B, C and D), all these 80 putative TA genes are represented in 4 circo plot to show the BLASTP results against the three TA carrying pathogens. In next step, we took all 80 putative TA protein sequences of *H. pylori* as 'Query Sequence' against the experimentally reported TA sequences of these three TA carrying pathogens. As a result, here we found HP0892 the toxin of *H. pylori* has the sequence similarities with a toxin of the YoeB-YefM TA system and the antitoxin HP0893 has similarities with the antitoxin for the HicAB TA system in *E. coli* K-12 substr. MG1655. The results

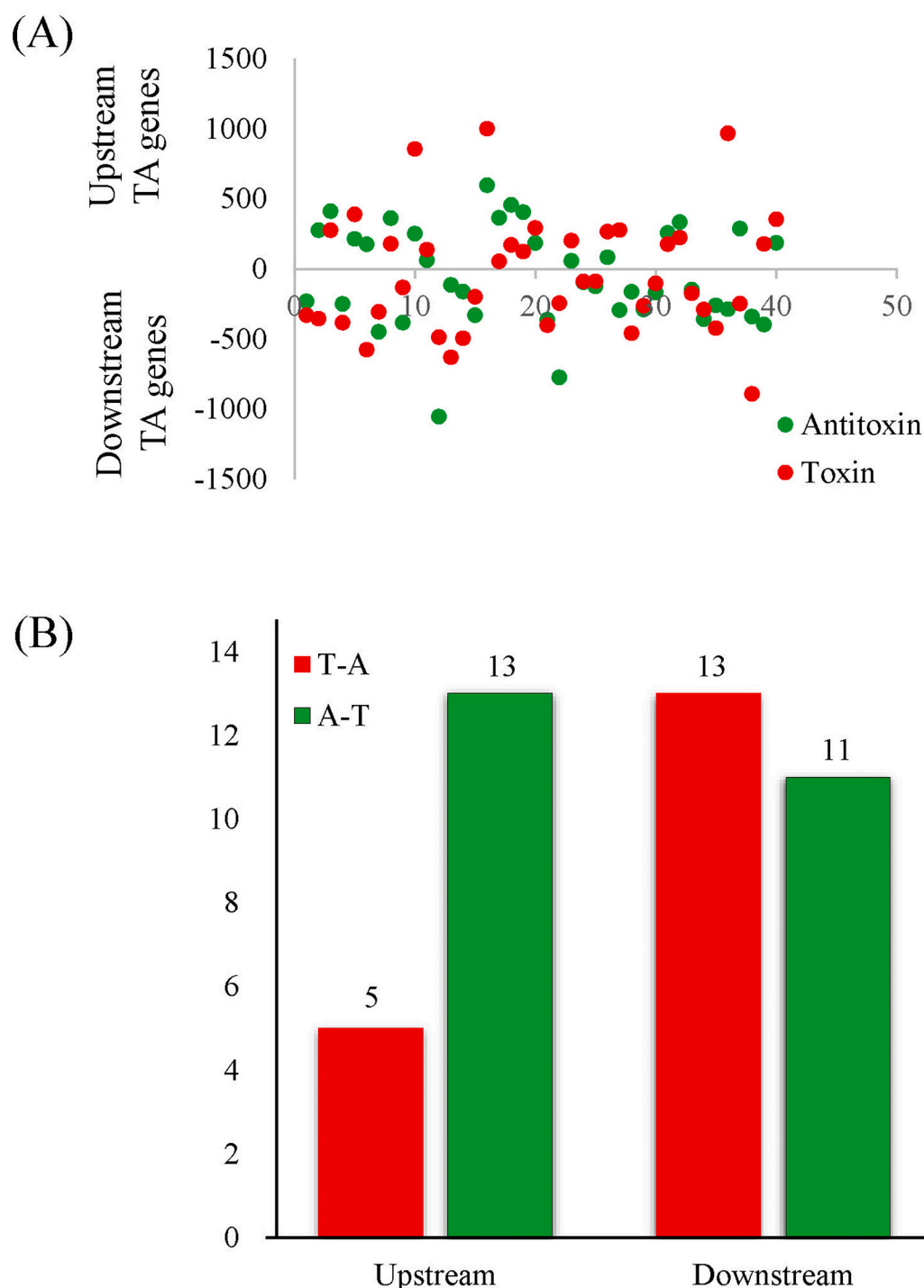


Fig. 5. Upstream and Downstream distribution of TA genes (A) Antitoxin (green dots) and Toxin (red dots) genes are shown in Scatter chart representing their protein length, length on positive scale (+ strand) and length on negative scale (– strand). (B) Arrangement of TA pairs in their operon models as A-T or T-A. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

provided the data related to the similarity of the TA genes identified against TA present in other pathogens.

3.7. Network-based analysis to find interlinked putative TA genes of *H. pylori* 26695

Protein-protein interactions (PPIs) or Protein network studies are of great use to find out the interaction of toxin-antitoxin genes in a TA pair and their function in a particular TA system. As we know PPIs plays an important role in cell physiology in both normal and diseased

conditions, each PPI possesses specific biological functions. A network-based analysis can be taken into consideration to understand completely the TA genes associations and their biological phenomena. We used an online-based classification system STRING (version 11.0) (<https://string-db.org/>) to find the associated toxins to their antitoxins and vice versa. STRING is an open-source online database which implements KEGG and Gene Ontology classification system [37]. This database is getting new updates regularly and it includes total 5090 organisms, 24.6 million proteins and more than 2 billion interactions. We used this database for our PPIs studies and input all 80 putative TA

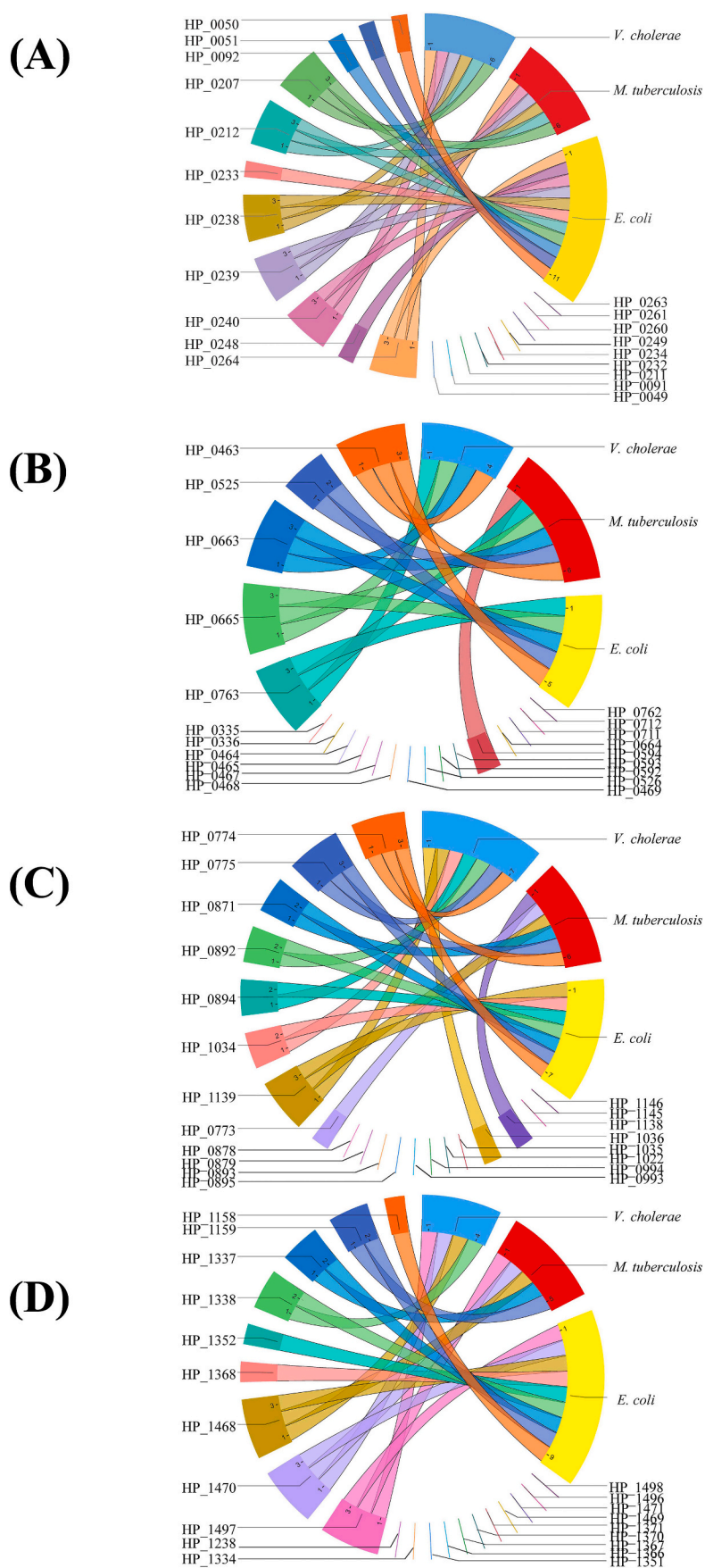


Fig. 6. Circos plots showing the BLASTP results filtered by query cover >50% (A), (B), (C) & (D) results for all 80 putative TA genes have shown in 4 circos plots, each plot showing 20 TA genes).

genes identified by TASmania individually to get the linked genes for each input. The output network-based interactions were displayed with different categories and each category was represented by seven different coloured lines. Red line indicates the presence of gene fusion evidence, green line indicates the neighbourhood evidence, blue line indicates gene co-occurrence, purple line indicates evidence related to experimental analysis, yellow line indicates text mining evidence, light blue line indicates the database evidence, and black line indicates the co-expression evidence. This database is giving a combined score for the predicted interaction partners by which the closely relative protein can be detected for the input gene. Based on the combined score, the interacted partners were displayed from high to low confidence, where 0.9 is the highest confidence, 0.7 is high confidence, 0.4 is medium confidence and 0.15 or better are considered as low confidence. By STRING database out of 80 putative TA genes, we found the networks for 74 genes and six genes i.e. HP0335, HP0336, HP0663, HP0712, HP1145, and HP1146 were not present in this database (Supplementary Table S8). For example, when we searched the interaction for all reported TA genes HP0892, HP0893, HP0894, and HP0895 we found that they are interlinked with each other and vapD protein as shown in the Fig. 7A and when we searched for four putative TA genes HP0993, HP0994, HP1337, and HP1338, we found that HP0993 is interlinked with HP0994 and HP1337 is interlinked with HP1338 as shown in Fig. 7B. Both HP0994 and HP1337 are interlinked to nadC protein and HP1337 belongs NikR protein family. According to our previous Pfam analysis we found that NikR protein family experimentally found as a toxin family in *E. coli* [59,60].

To input multiple proteins simultaneously and visualize the combined networks for all inputs in a single frame we used an open-source application Cytoscape (version 3.8.2) (<https://cytoscape.org/>) which is implemented on Java (version 11.0.6) [38]. This application is freely available for different operating systems, and it is integrated with the STRING database and some other popular automation tools like Python and R studio. So, in our output large networks, the individual genes are interlinked with each other by different coloured lines to specify different categories and the thickness of line showing low to high confidence between two interlinked genes (Fig. 7C).

4. Discussion

The present study exhibited the discovery and mining of TA genes from *H. pylori* 26695 genome using *in silico* databases and tools. Initially, a newly developed TA database was utilized to search genome wide TA in *H. pylori* which revealed the presence of enormous no. of TA gene i.e., 80 in this human pathogen (Supplementary Table S1). This is known that *H. pylori* is not well explored yet for the presence of TA genes in its genome despite being a persister and also make biofilm. However, *M. tuberculosis* and *E. coli* have been thoroughly studied and characterized for the presence of TA genes and their role in persistence, as stress managers and biofilm formation [7,8]. Hence, this study was validated by comparing the putative TA genes from other three known TA carrying pathogens i.e. *M. tuberculosis*, *E. coli* and *V. cholerae*. This comparative analysis can strengthen the data obtained for not much explored pathogen *H. pylori* for TA genes. Further, the TA database has also given the information related to the protein families of putative genes. The identified protein families were also compared with their experimental evidence in another three well studied TA carrying pathogens. Total 17 protein families were present in the *H. pylori*, it is interesting to know that 12 of them were experimentally verified and present in *E. coli*, 5 out of 17 is known to be present in *M. tuberculosis* and 3 out of 17 in *V. cholerae* (Table 1). However, 5 of the putatively identified protein families of *H. pylori* were not found in these three organisms (Fig. 4). Upon search, for their presence in another pathogen, 1 is found in *B. subtilis* [70] and 1 family has the RelE homologous domain which belongs to the Type II TA [71–73]. It is worth noted that a large no. of identified protein families of *H. pylori* were belongs to TA families

because either the toxin or antitoxin gene sequence carries the domains similar to TA systems. On the other hand, it is identified from the literature search that 7 families of *H. pylori* are matching with the Type II TA (Table 1) modules while 2 are showing similarity with Type IV and 1 with Type I. For e.g., PhdYefM, belongs to Type II module and interestingly overlapping in all three example pathogens. The antitoxin of this family binds to the partner toxin and the complex subsequently undergoes to unfolding with the increase in temperature [61–63]. One of the experimentally identified *H. pylori* TA set i.e., HP0892 and HP0893 have revealed that HP0892 i.e., a toxin binds strongly to HP0893 antitoxin in a 1:1 stoichiometric ratio where toxin matches at the sequence level or showed homology with YafQ toxin having intrinsic RNase activity [25]. Another experimentally verified TA pair toxin i.e., HP0894 also belongs to this YafQ family [24]. This toxin family is also present in *E. coli* and *V. cholerae* [46,65,66]. It has also been known that this toxin family participates in the biofilm formation and also increases the persister cell formation [66,75,76]. This gives an insight and clue that the putatively identified TA genes in the *H. pylori* may plays role or participates in these complex processes such as persister cell formation, biofilm formation and antibiotic resistance. Therefore, it is worth identifying and working on the TA systems in human pathogen *H. pylori* 26695.

In parallel studies, the putatively identified TA genes were also validated by studying the linked genes using a tool called SLING [33] which is known to identify the linked genes of an operon. Further, different operon-based analysis was performed using several databases for e.g., Microbes online Prokaryotic Operon database [34,40] and DOOR 2.0 [36] known for the operon-based analysis of the genes. All this study reveals the presence of the putative TA genes identified by TASmania database [41] and the details are given in the specific result section. Our results were also in line with the experimentally proved study performed for operon-based transcriptome studies of *H. pylori* [74]. Additionally, network-based studies were also implemented using String network analysis to identify the interlinked putative TA genes (Fig. 7A, B and C).

Identification of bonafide TA pairs and subsequently their characterization will perhaps shed the light on their role as stress managers because their presence on chromosome may improve the fitness of pathogen in different stress condition [77]. As per our understanding, this is the first report in *H. pylori* 26695 revealing the presence of enormous no. of TA widely distributed on its genome. Though, the identification of presence of large no. of TA genes, types and families are still incipient and important to reinforce that this is based upon *in silico* results and need to check during further studies. With no doubt, special attention is required to characterize putatively identified TA genes also to know the regulatory network and their role in human pathogen *H. pylori*.

Competing financial interests

The authors have declared no competing interest.

CRediT authorship contribution statement

Nidhi Gupta: Conceptualization, Methodology, Validation, Data curation, Writing-original draft, Writing-reviewing and editing, Project administration. **Deepak Kumar Behera:** Methodology, Investigation, Data curation, Writing-original draft. **Vijay Kumar Prajapati:** Conceptualization, Resources, Supervision. **Vijay Kumar Verma:** Conceptualization, Methodology, Writing-reviewing and editing, Supervision, Funding Acquisition.

Declaration of competing interest

Authors wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant

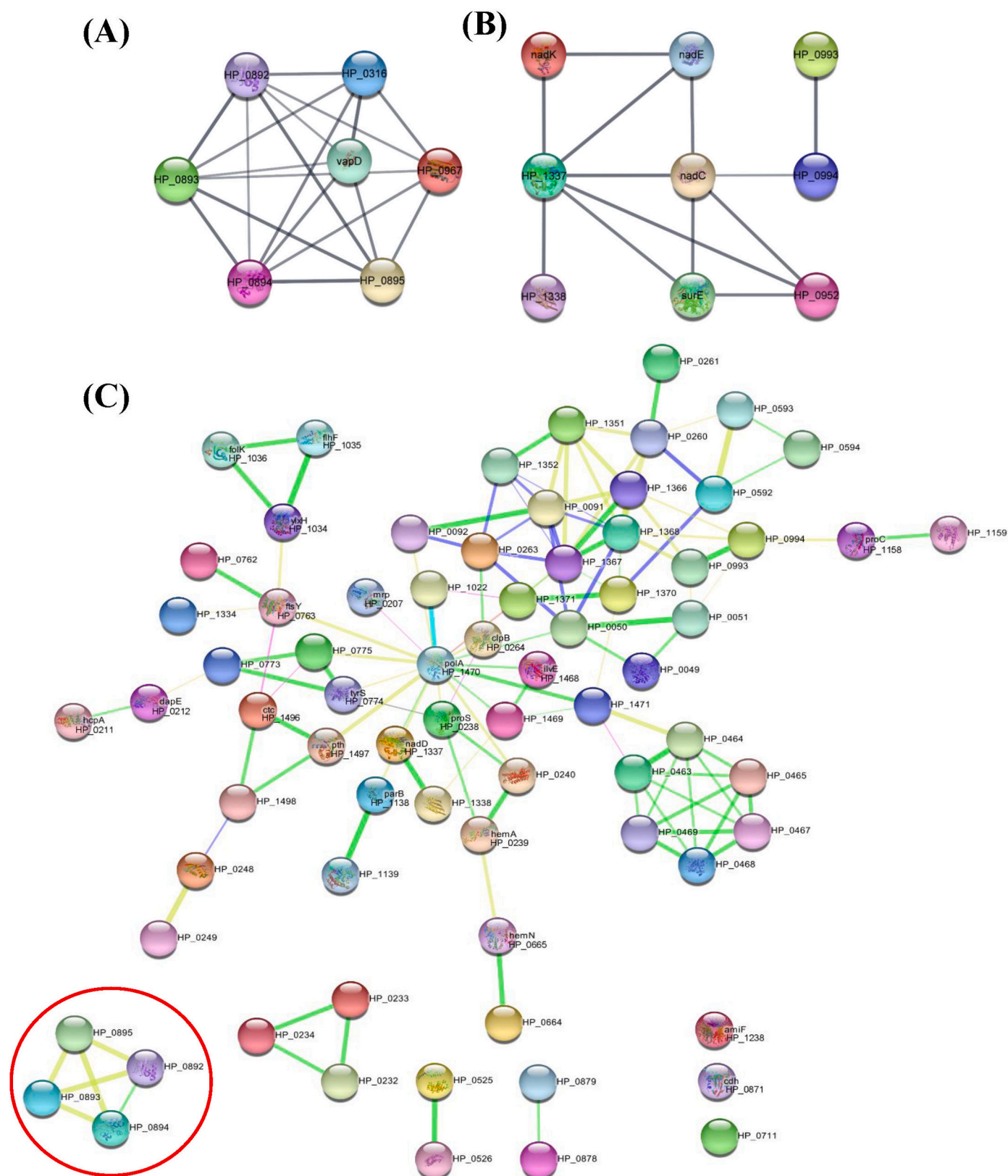


Fig. 7. Network showing interlinked TA genes (A) Network showing the interaction between reported TA genes HP0892, HP0893, HP0894, and HP0895 (B) Network showing the interaction between putative TA genes HP0993, HP0994, HP1337, and HP1338 (C) Network showing 74 putative TA, available in the STRING database (all putative TA genes are given as input in Cytoscape to draw the entire network, already reported TA interlinked genes are marked in a red circle and the noninteracting putative TA genes are shown alone as they are not linked with any putative TA genes but they are interlinked with other genes present in *H. pylori*). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lfs.2021.120149>.

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